Reliable LLM Issue Triage Through Calibrated Deferral and Production Testing

UIC CS544 Machine Learning in Production | Hybrid project report

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Research problem. An issue-labeling service must decide whether to assign bug, feature, or question automatically or ask a maintainer to review the issue. We study how much automation is possible at a target error rate, and whether that decision remains reliable as repositories, time periods, and writing styles change.

Background. Prior work studies classical and LLM issue classification [1,2]. Temperature scaling, selective classification, risk control, and conformal prediction offer ways to assess confidence and defer uncertain cases [3-6]. Behavioral testing and research on user-related performance differences motivate controlled wording tests [7,8]. Explanation evaluation and production testing inform supporting analyses [9,10].

Goal and scope. The contribution is a reproducible evaluation of selective issue triage, including operational tradeoffs, controlled behavioral tests, and regression checks. The core study concerns automatic-label errors and human-review workload; explanations and monitoring provide supporting diagnostics.

Research questions

RQ1. How do classical and LLM triagers compare in accuracy and cost across benchmark, recent, and held-out-repository issues?

RQ2. How much coverage can deferral policies achieve at a 10% error target, and how does target attainment change under shift?

RQ3. How do controlled changes in wording and reporter cues affect predictions and referral to human review?

RQ4. Do quoted explanations identify evidence that influences the classifier's decisions?

RQ5. Which signals detect changes in issue traffic, and which service faults do regression tests catch?

Working hypotheses. H1: LLM triagers transfer across repositories better than classifiers trained on issue history. H2: Confidence-based deferral reduces automatic-label errors while sacrificing coverage; changes in traffic and reporter style affect both reliability and review workload.

General approach. Using matched issue text, compare TF-IDF and embedding logistic regression with local Qwen2.5-7B in zero-shot and few-shot modes; the latter retrieves six labeled examples. Evaluate auto-all, confidence thresholds, risk-controlled thresholds, singleton conformal sets, and classifier agreement. Planned validation separates temperature fitting from certification and excludes target repositories from training and calibration. Human annotators will check rewrites; reporter-cue controls will separate newcomer status from question-like wording. Supporting tests compare evidence deletion with matched random deletion and evaluate monitoring on independent clean traffic.

Evaluation benchmarks and metrics. Use NLBSE'24's 3,000 issues from five repositories: 1,005 history, 495 reserved for calibration stages, and 1,500 test. The recent collection contains 1,430 single-type issues from 2025-26, with 514 sampled for evaluation. Reuse benchmark subsets for held-out-repository testing (750 issues) and behavioral probes (210). Measure accuracy, macro-F1, calibration error, coverage (share auto-labeled), and selective risk (errors among automatic labels), with uncertainty intervals. Also measure rewrite-induced label and referral changes, quote validity and deletion effects, monitoring delay and false alarms, mutation detection, and end-to-end latency.

Selected literature and project relevance

References 1-2 cover issue classification; 3-6 support calibration and deferral; 7-8 motivate behavioral and fairness tests; 9-10 guide explanation and production testing. Paper titles link to primary sources.

[1] Kallis, R., Di Sorbo, A., Canfora, G., and Panichella, S. (2021). Predicting Issue Types on GitHub. arXiv:2107.09936, preprint.
Task foundation: presents Ticket Tagger for issue-type prediction; motivates automated labeling and classical baselines.

[2] Aracena, G., et al. (2025). Applying Large Language Models to Issue Classification: Revisiting with Extended Data and New Models. arXiv:2506.00128, preprint.
LLM baseline: evaluates issue classification across NLBSE datasets; provides context for our deployment-focused study.

[3] Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. ICML 2017.
Confidence calibration: supplies temperature scaling and calibration-error analysis for classifier scores.

[4] Geifman, Y., and El-Yaniv, R. (2017). Selective Classification for Deep Neural Networks. NeurIPS 2017.
Selective automation: motivates abstention and evaluation through the tradeoff between risk and coverage.

[5] Angelopoulos, A. N., et al. (2021). Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. arXiv:2110.01052, preprint.
Risk control: supplies hypothesis-testing procedures for certifying an error target using held-out calibration data.

[6] Angelopoulos, A. N., and Bates, S. (2021). A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv:2107.07511, tutorial paper.
Prediction sets: motivates singleton-set automation and states the exchangeability assumptions behind coverage guarantees.

[7] Ribeiro, M. T., Wu, T., Guestrin, C., and Singh, S. (2020). Beyond Accuracy: Behavioral Testing of NLP Models with CheckList. ACL 2020.
Behavioral testing: guides controlled invariance tests for wording, template structure, and reporter cues.

[8] Poole-Dayan, E., Roy, D., and Kabbara, J. (2024). LLM Targeted Underperformance Disproportionately Impacts Vulnerable Users. arXiv:2406.17737; accepted at AAAI 2026.
Fairness motivation: studies response-quality differences by user traits, including English proficiency; motivates style tests.

[9] DeYoung, J., et al. (2020). ERASER: A Benchmark to Evaluate Rationalized NLP Models. ACL 2020.
Explanation testing: motivates rationale deletion tests and separate evaluation of plausibility and faithfulness.

[10] Breck, E., Cai, S., Nielsen, E., Salib, M., and Sculley, D. (2017). The ML Test Score: A Rubric for ML Production Readiness and Technical Debt Reduction. IEEE Big Data 2017.
Production readiness: guides monitoring, regression suites, and tests of operational failure paths.